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Solar Irradiance Forecasting using Improved Sample Convolution and Interactive learning

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Abstract

Renewable energy sources, driven by the global concern for climate change and the urgent need to reduce greenhouse gas emissions, are gaining increasing importance. Among them, solar energy stands out as a crucial renewable source and accurate solar irradiance forecasting is vital for its effective utilisation. Deep learning (DL) techniques have emerged as state-of-the-art solar irradiance forecasting methods in past years. However, traditional models that are widely used have inferior accuracy in forecasting. To bridge this gap, this paper presents an improved state-of-the-art model that leverages the convolutional property to forecast the time series data. We got a reduction of 8.32% in MAE compared to the base model. The paper's findings highlight the potential of DL-based approaches to significantly improve the accuracy of solar irradiance forecasts.

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Keywords: Solar irradiance; NWP models; Power scheduling; Deep learning; time series; forecasting; Convolutional Neural Networks; NSRDB

1. Introduction

Energy consumption has steadily increased over the years, driven primarily by fossil fuels like coal, oil, and natural gas. However, these non-renewable sources are finite and pose significant environmental risks, such as air pollution and greenhouse gas emissions. This has led to a global push towards more sustainable and renewable energy sources like solar, wind, hydro, geothermal, and biomass. Among these renewable sources, solar energy is one of the most rapidly developing and popular options due to its easy installation and almost unlimited supply.

In addition to these benefits, solar energy offers other advantages. For example, solar power is more reliable than other renewable sources, such as wind or hydro, as it is not dependent on wind or water availability. Additionally,

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solar panels can be installed on various surfaces, from rooftops to large-scale solar farms, making it a flexible and scalable option for power generation. To fulfil the increasing need for solar power, advancements in technology have led to improved solar panels' efficiency, reducing installation and maintenance costs. Solar energy has become more accessible to individuals and communities, and governments worldwide are investing in large-scale solar projects to support sustainable power generation.

Two primary mechanisms for generating solar power are photovoltaic cells and solar thermal power plants. In both cases, the amount of solar irradiance, or the amount of sunlight reaching a particular location, plays a critical role in how much power can be generated. Because solar irradiance is unpredictable, accurate forecasting is necessary to schedule and efficiently dispatch power production. This is where solar irradiance forecasting comes in. Power production can be planned, and dispatching schemes can be formulated by accurately predicting how much solar irradiance will be available at a given location. This can help ensure enough power is produced, even on cloudy or less sunny days, and alternative sources are utilised if needed.

2. Literature Review

The search methodology for this literature review was conducted using the PRISMA model (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) [19]. A systematic search of relevant studies was conducted in several scientific databases, including MDPI, IEEE-Xplore, ScienceDirect, etc. The search was limited to studies published in English language. The following search terms were used: "solar irradiance forecasting", "time series forecasting", "solar irradiance forecasting using deep learning", and "electric load forecasting". Additionally, we reviewed the reference lists of the retrieved articles to identify any relevant studies that may have needed to be included in the initial search.

Figure 1 shows the process for selecting articles based on different screening levels. The initial search yielded 350 articles. From this, duplicates were removed and screened based on the title first and again based on the abstract to identify relevant studies for full-text review. The majority of the research work uses solar irradiance data, which is

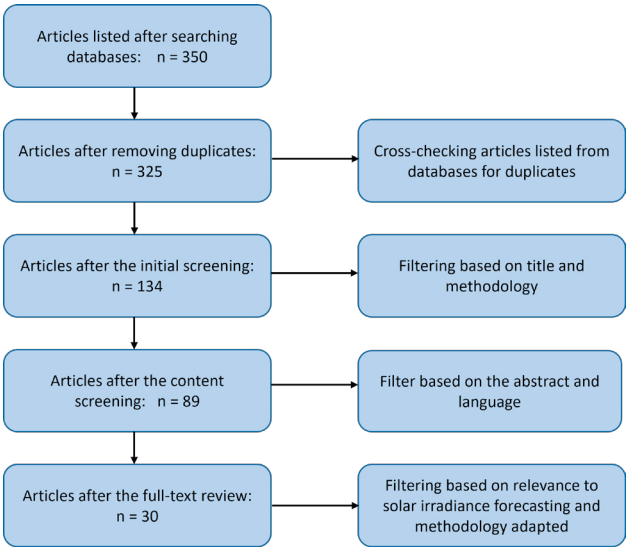


Fig. 1: Article selection criteria - PRISMA Diagram

time series data sequentially taken at successive time instances. A few published research works [13][27] also use sky images to forecast solar irradiance. The following subsections detail a comprehensive review of the published research articles by categorically placing the research works based on the models and the data type used.

- Image-based Forecasting: In image-based methods, satellite images [23] are used for solar irradiance forecasting. Out of the 30 papers reviewed, 17% have used only images, and 83% have used only numerical data.

Among the methods that used image data, 40% used satellite images, and 60% used sky images. Multiple strategies are followed in the forecast. Convolutional models or Deep learning models are used for feature extraction predicted using SVM [17], ensemble [22] methods. Cloud cover data [27] can be extracted from sky images, which can be used for forecasting.

However, Image-based methods require significant computational resources for image processing and feature extraction, which can be time-consuming and computationally expensive. These methods may struggle to accurately forecast solar irradiance under certain weather conditions, such as overcast skies or rapidly changing cloud cover.

- Time series data-based Forecasting: Because of the easy availability of numeric time series data, research on forecasting based on time series data have more attention than satellite image-based forecasting. The data may be collected from ground stations [30] and satellite data providers[9]. Many countries provide weather parameters online. The availability of data made a huge impact on the advancements in this research area. NASA's POWER project (https://power.larc.nasa.gov/data-access-viewer/), NSRDB: National Solar Radiation Database (https://nsrdb.nrel.gov/data-viewer), etc. are some freely accessible sources for solar irradiance data.

2.1. Based on the model used for forecasting

There has been a huge development in the models used for forecasting in the recent past. Nowadays, deep learning-based methods report the best accuracies in time series predictions. The following subsections provide a comprehensive survey of the popular models used for forecasting.

2.1.1. Numerical weather forecasting models

Numerical weather forecasting models use mathematical models for forecasting. Hadrien et al. [32] utilised numeric weather forecasting methods and statistical analysis to predict solar irradiance in tropical areas. They employed the Weather Research and Forecasting (WRF) model and combined it with multivariate statistical learning techniques for forecasting. WRFsolar and their proposed statistical learning method outperformed other forecasting methods, including smart persistence [18], a climatological forecast, and GFS for day-ahead irradiance forecasts. This study highlights the potential of integrating advanced modelling and statistical analysis techniques to improve solar irradiance forecasting in tropical regions.

Arca et al. [5] used a numerical weather prediction model for reference evapotranspiration prediction. They analysed the evapotranspiration calculated using analytical models like Penman-Montaith [15] and NWP (Numerical Weather Prediction) models. The results were compared with ARIMA [8] and ANN models, and they got better results from NWP models. Dynamic Mode Decomposition(DMD) [2] [4] [24] is another method which recently shown promising results in time series forecasting

As mentioned in the last paragraph, there is a lot of research on weather forecasting using NWP models. However, the main problem with these models is their limited spatial and temporal resolution, and they may not accurately capture small-scale variations in solar irradiance. Also, they can have biases in their output, which can lead to errors in solar irradiance forecasting. These models are computationally intensive and require significant computational power and running time, which makes them less practical for real-time forecasting applications.

2.1.2. Machine-learning models

Recently, ML models, particularly the DL methods, have produced state-of-the-art results on prediction [34] [28] [29] and classification [31] problems. The new trend is applying ML methods to the solar irradiance forecast. The following subsections review the ML-based models in the solar irradiance forecast.

- Traditional ML models: In machine learning-based models, linear models were used for forecasting solar irradiance because of the model's simplicity and easy-to-understand features. Autoregression(AR) [35], Moving average (MA), autoregressive moving average (ARMA), autoregressive integrated moving average (ARIMA) [30], seasonal autoregressive integrated moving average (SARIMA) [20], etc. are the main linear models that have been used. These models, in the given order, showed improvements in reducing the error in prediction. However, the models cannot understand the long-term seasonality and trend in the solar irradiance

data, which makes these models less suitable for practical applications.

- Neural Network Models: NN-based models[25] were also used in past years to overcome the disadvantages of linear models. Neural networks are capable of understanding the complex behaviour of time series data. They are capable of remembering the knowledge as they go through the data. This makes neural network-based models a good choice for time series forecasting. Van bui et al.[10] used RNN for forecasting one hour-ahead load. They proposed a multi-scale RNN architecture for the same. ANN was used for applications like Hydrological drought forecasting [1], peak load forecasting [26], etc.

- Advanced RNN models: As neural networks advanced, deep learning models were able to satisfy forecasting needs. Different RNN architectures like LSTM, GRU, Bi-LSTM, Bi-GRU, etc., were used for different applications like covid-19 forecasting [6] predicting energy consumption [3], solar irradiance forecasting [16], [14] etc. Aslam et al. [7] used GRU for hourly and daily solar irradiance forecasting for the year ahead horizon. Compared with other models like LSTM, RNN, and FFNN (Feedforward neural networks), GRU outperforms other models in studying the complex structure of solar irradiance data, mainly because of the large data size.

- Hybrid models: Wang et al. [33] proposed a new approach from conventional methods that use a manual process for feature extraction, a traditional technique for modelling, and single DL models, which may fail to find the complex behaviour of data. So, they use Discrete Wavelet Transformation (DWT) based solar irradiance sequence decomposition, a CNN-based local feature extractor, and an LSTM-based sequence forecasting model. They are also considering the dependency of weather on solar irradiance. Dairi et al. [12] proposed a model by combining the favourable qualities of the Bi-LSTM and attention mechanism with the VAE model, a unique deep attention-driven model for solar irradiance forecasting.

  Brahma et al. [9] have a different approach that uses multi-site data, i.e., the dataset contains the GHI value of the target location and the GHI of nearby locations. From the data correlation of nearby areas, GHI is found in the target location, and the locations with high correlation are taken as parameters for forecasting.

From this research, it is evident that numerical weather forecasting and image-based forecasting have some disadvantages in terms of computational complexity and difficulty in acquiring the data. Also, in recent years, Deep-learning-based models have gotten colossal attention. Even though there are both image-based and numerical-featured data-based forecasting, the latter has gained more momentum. Also, the research shows that most of the focus is on univariate forecasting, considering Global Horizontal Irradiance only. Even though meteorological factors like temperature, solar zenith angle, clouds, etc., impact GHI significantly, these parameters are not considered in many research articles.

In this research article, we put forward an improvement on one of the state-of-the-art deep learning model [21] for solar irradiance forecast. We experiment with the architecture both in univariate and multi-variate settings. With the proposed architecture, we could achieve a reduction of 8.32% in MAE. The remaining part of the paper is organised as follows. The data collection and methodology are detailed in section 3. The results and comparisons are shown in section 4. Finally, concluding remarks and future perspectives follow in section 5.

3. Methodology

3.1. Data Collection and parameter selection

The model forecasting solar irradiance and related meteorological parameters uses time-series data. However, collecting data at the desired locations using ground-based stations can be difficult, and the non-availability of such data is a common problem. To overcome this issue, satellite-based data is used, which is collected from the National Solar Radiation Database (Figure 3). The data is collected for the location Nedumbassery, Kerala, India where the world's first airport fully powered by solar energy, CIAL is situated. This database offers a rich source of meteorological parameters, such as GHI (Global Horizontal Irradiance), DHI (Diffuse Horizontal Irradiance), DNI (Direct Normal Irradiance), Ozone, Solar Zenith Angle, Cloud Type, Relative Humidity, Pressure, Wind, etc. While these

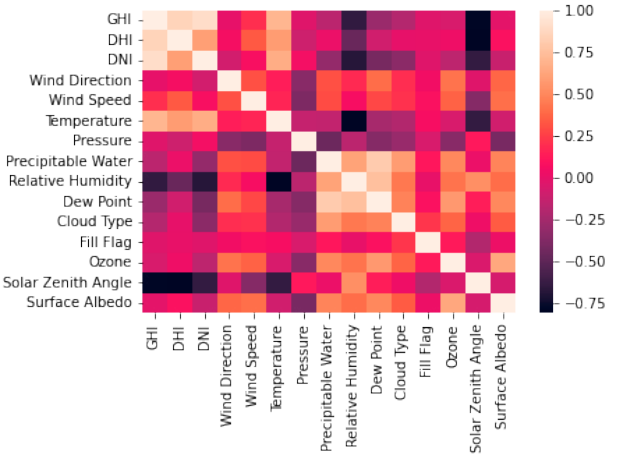


Fig. 2: Correlation matrix of data

meteorological factors significantly impact solar irradiance, not all parameters contribute equally to modelling. Thus, it is necessary to find the correlation between each parameter and solar irradiance and select the ones that show the strongest correlation. After analysing the correlation matrix (Figure 2), it was observed that DHI, DNI, Temperature, Precipitable Water, relative humidity, and solar zenith angle correlate more strongly with GHI or with other weather parameters. Some parameters are dropped because of their nonchanging behaviour. Hence, only these seven parameters are considered in forecasting, as they provide essential information about solar irradiance. This approach has potential applications where accurate and reliable solar irradiance forecasting is essential, such as renewable energy, agriculture, and research related to 'climate change'.

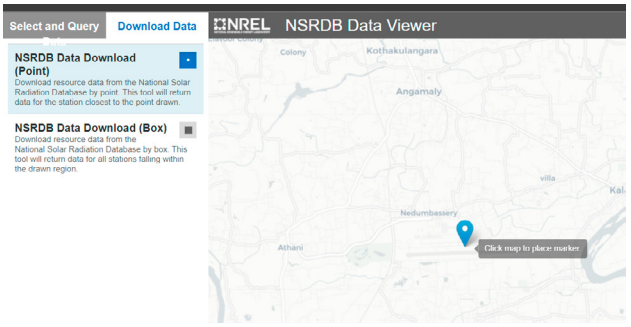


Fig. 3: Data point selection adapted from the National Solar Radiation Database (https://maps.nrel.gov/nsrdb-viewer/)

3.2. Forecast Methodology

The forecasting is done in both univariate and multivariate settings. In the univariate setting, only the target location's Global Horizontal Irradiance (GHI) value is input to the model. In contrast, other weather variables, such as DHI, DNI, Temperature, Precipitable Water, relative humidity, and solar zenith angle, are also included as input in the multivariate setting. These parameters were selected from the correlation matrix analysis (Figure 2). Additionally, a multi-site data setting has been experimented with to improve the forecasts' accuracy. In this setting, the GHI values of the target location and the nearby area's GHI values are considered together to improve the prediction. This approach considers that weather conditions can be highly correlated between neighbouring sites.

For this research, the target location is Nedumbassery, a place in Ernakulam district, Kerala, India. Cochin International Airport Ltd (CIAL) is in Nedumbassery and is the world's first airport fully powered by solar energy. CIAL

started the solar project in 2013 with an initial capacity of 100 kWp, and by 2015, the airport had an installed capacity of 13.1 MWp. CIAL now has a total solar installed capacity of 50Mwp. In green energy generation, CIAL has a significant impact. Because of this, Nedumbassery is one of the best choices for the target location. Based on this location, datasets have been created for univariate, multivariate, and multi-site data settings. Considering these datasets as a benchmark, we have tried popular CNN and RNN-based deep learning models. We have experimented with state-of-the-art models like LSTM, GRU, Bi-LSTM, and SCINet. The results are summarised in Table 1. This shows that SCINet is the best performer among the models, and prediction based on multi-variate data yielded a reduction of 78.86% in MAE. Then, we proposed enhancement to the SCINet models, which has produced state-of-the-art results (refer to section 4). The following subsections briefly explain the popular Deep-learning architectures used in this study.

3.2.1. Proposed Architecture

Recently, CNN-based networks [21] reported excellent results in temporal and spatiotemporal datasets such as traffic [11], electricity transformer temperature [36], and exchange rate. The SCINet [21] used convolution to capture time series close correlations at multiple levels and then use these features for prediction. The interleaved samples of the time series data hold the same relationships. Convolutions can be used to capture this relationship and when these features interact, these turn out to be excellent feature series in time series prediction. In the original architecture, the convolution block consists of two 1D convolution layers and activation layers, including leakyReLU and Tanh. SCINet provides several advantages over conventional convolutional architectures by extending the receptive area of the convolution procedure, allowing for multi-resolution analysis, and enabling the extraction of temporal connection characteristics more predictably.

The SCI-Blocks are at the heart of SCINet, which are arranged in a binary tree structure. The SCI-Block is the basic building block of the SCINet, and it contains convolution layers. When input data enters the SCI-Block, it is divided into odd and even series, processed separately through blocks containing convolution layers and exponential modules, and then multiplied with the opposite data. The processed odd series is then combined with the processed even series, and vice versa, using an interactive layer to overcome the losses due to downsampling. CNN serves as a pivotal component in our model. CNN helps to extract the local patterns and spatial hierarchies within the data. The features extracted by the CNN are subsequently utilised in conjunction with other layers within the architecture to accomplish the overarching objectives of our study.

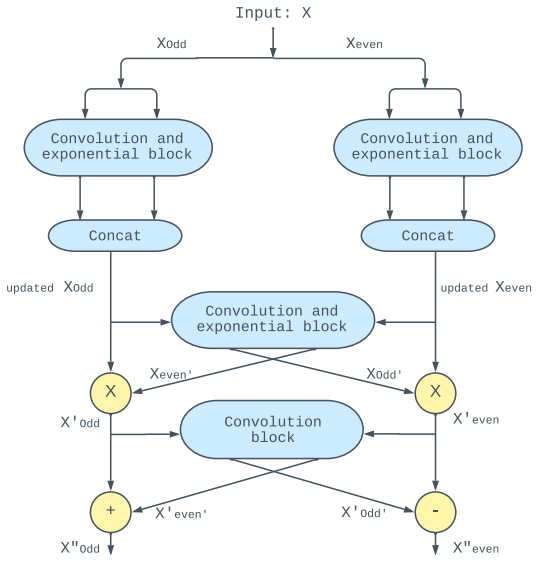


Fig. 4: Improved SCI-Block

The most groundbreaking aspect of our modification lies in the concept of multilevel interaction. We've taken a significant step forward by not only splitting time series data into odd and even sequences but also allowing them to interact at multiple levels. This innovative approach revolutionises the way we understand and model temporal data, enabling intricate cross-level relationships that unveil hidden patterns and insights. This novel approach leads to a deeper and more nuanced understanding of the intricate patterns within the data. Figure 4 illustrates this process, showcasing how the interaction between these distinct sequences refines our grasp of the data's intricate features and enhances our ability to capture underlying patterns and trends.

One of the pivotal advancements in this modification is the replacement of the leakyReLU activation function with SiLU. This change is not merely a matter of preference but a well-founded decision that enhances our model's performance on this specific dataset. SiLU, with its unique characteristics, proves to be more adept at capturing the intricacies of our data. It provides superior results (Table 2), particularly when dealing with the complexities presented by our dataset.

Incorporating these improvements, the modified SCI-Block unlocks a more profound understanding of input data features, offering superior accuracy in forecasting. The SCINet architecture, empowered by this modification, emerges as an invaluable tool for modelling temporal data across various application domains.

4. Results and Comparisons

This study conducted experiments on different data sets and machine-learning models to forecast solar irradiance. It also aimed to compare the performance of different models and architectures and propose improvements to increase forecasting accuracy. The evaluation metrics used in the experiments are mean absolute error (MAE), and root mean square error (RMSE). MAE is easier to interpret because it represents the average absolute deviation from the true value and is less sensitive to outliers than RMSE. Table 1 shows the forecasting results on different models which were done multiple times. The final value shown is the mean and standard deviation of multiple test results, which helps to understand the reliability of the models.

The four data settings, i.e., univariate, multivariate, multisite-univariate, and multisite-multivariate, are tested with LSTM, GRU, Bidirectional LSTM, and SCINet. The results showed that the SCINet model outperformed all other models. This experiment led to the base model selection as SCINet. Also, univariate showed better results than multivariate in meteorological and multi-site datasets, indicating that incorporating nearby areas GHI did not improve the forecasting accuracy. However, in the case of SCINet, the model could leverage meteorological factor dependency and provide better results in the same setting. So, the remaining experiments were focused on SCINet with multivariate settings considering DHI, DNI, Temperature, Precipitable Water, relative humidity, solar zenith angle, and GHI as parameters.

After hyper-parameter tuning, SiLU performed better than leaky ReLU, which was used inside the convolution block. To learn the underlying temporal characteristics of data more efficiently, adding convolutional blocks and interactive learning can help. These blocks are added after normal sci-block before it goes to the next level. This modification showed that the performance improved with more interactive learning blocks (Table 2). SCINet 1.0

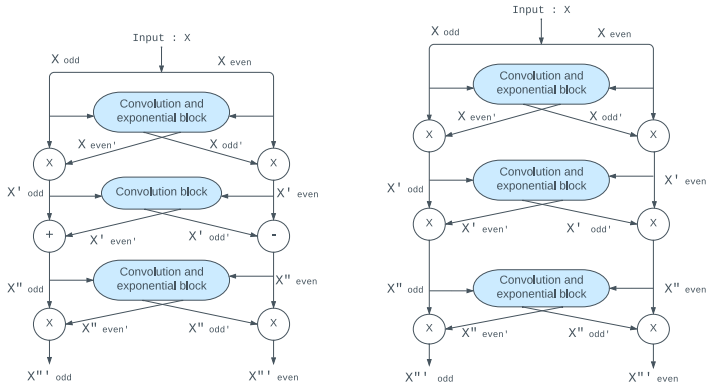
Table 1: Forecasting Results

		Univariate Setting	Multivariate Setting	Multisite-univariate setting	Multisite-Multivariate setting
HMM	MAE	213.36+-14.29	78.18+-11.46	202.62+-5.79	184.10+-27.10
	RMSE	345.91+-55.08	156.30+-12.57	355.44+-45.87	325.41+-48.47
LSTM	MAE	76.57+-2.96	129.85+-22.09	97.41+-6.02	148.59+-3.73
	RMSE	110.68+-4.40	202.91+-35.50	139.97+-5.18	234.22+-7.90
GRU	MAE	75.32+-4.21	127.21+-6.66	88.80+-14.12	155.55+-5.84
	RMSE	108.49+-7.06	187.04+-17.36	127.59+-13.83	242.88+-12.35
Bi-LSTM	MAE	73.86+-5.42	98.36+-3.12	81.73+-5.02	135.49+-6.85
	RMSE	106.56+-6.66	128.73+-2.41	120.97+-6.28	215.28+-11.64
SCINet	MAE	**66.91**	**27.45**	**66.45**	**70.91**
	RMSE	122.52	73.63	128.19	141.83

Table 2: Forecasting results on multivariate data with different SCINet architectures

Forecasting Model	MAE	RMSE
SCINet	27.4532	73.6303
Tuned SCINet	27.4428	72.7694
SCINet 1.0	26.3866	72.2878
SCINet 2.0	25.5254	73.7634
Proposed SCINet 3.0	**25.1673**	**72.5180**

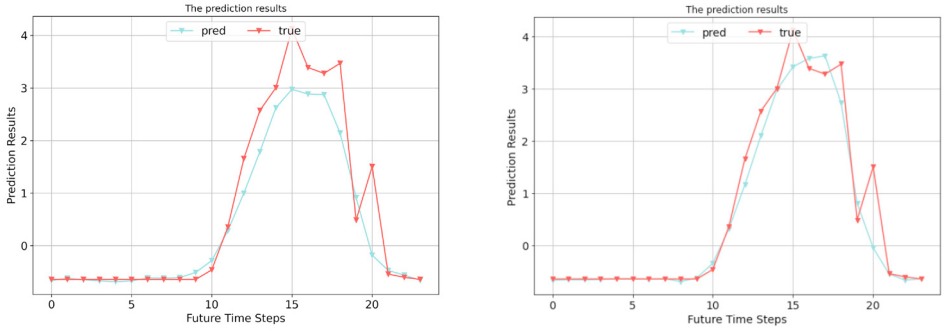
(Figure 5a) has an additional convolution-exponential block. In SCINet 2.0 (Figure 5b), the last convolution-only block is removed and added two convolution-exponential blocks. Even though SCINet 2.0 has a high RMSE, low MAE shows that the average difference from the actual value is much reduced.



(a) SCINet 1.0 (b) SCINet 2.0

Fig. 5: SCI-Block Architectures

The proposed architecture performed a multilevel convolution inside the SCI block(Figure 4), which extracts the deep features from the already midleveled samples. Before entering the normal SCI-Block, the incoming input data split into odd and even series. After going through the interactive learning layer constituting the convolution-exponential layer, this data concatenated back to the original series. Then, the data will pass to the normal SCI-Block. This architecture outperformed all other models and architectures tested, as indicated by the result reported in Table (2). The proposed architecture could produce an 8.32% reduction in MAE compared to the base model. This indicates that the proposed architecture has the potential to achieve more accurate solar irradiance forecasting and can be used for practical applications in renewable energy forecasting systems.



(a) SCINet (b) SCINet 3.0

Fig. 6: Results

5. Conclusion

This research paper sheds light on the importance of deep learning-based solar irradiance forecasting and puts forward an improved architecture over the state-of-the-art SCINet model for solar irradiance prediction. With the proposed modification, we could achieve an 8.32% reduction in MAE. Also, the paper gives a comprehensive review of different mathematical models and data used in solar irradiance forecasting. These methods have shown the potential to significantly improve the accuracy of solar irradiance forecasts, which can profoundly impact renewable energy planning and management. Altogether, this paper provides a valuable reference for researchers offering insights into recent developments, approaches, and challenges in solar irradiance forecasting. From the review, we learned that data quality, availability, and uncertainty often affect machine learning models' performance, leading to inaccurate predictions. Thus, improving data collection, pre-processing, and quality control is essential for advancing this field. Additionally, further research is needed to explore the use of hybrid models that combine various models, as well as the integration of other meteorological variables into the forecasting process.

The study establishes a groundwork for future research in solar irradiance forecasting, suggesting exploration into advanced forecasting models with emerging deep learning architectures and novel feature extraction techniques. Additionally, it recommends incorporating external factors such as climate patterns and geographical variations while emphasising the need to enhance the spatial and temporal scalability of the models for broader applicability across diverse geographic locations and forecasting timeframes.

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